Roll No. | | | | | | | |

EVEN END SEMESTER EXAMINATION – 2025

Name of the Course: B.Tech (ECE) | Semester: IV

Name of the Paper: Electromagnetic Field Theory | Paper Code: **TEC 404**

Time: 3 Hours | Maximum Marks: 100

Note:-

(i) All questions are compulsory.

(ii) Answer any two sub-questions among a, b & c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

Q1	(20 marks)	**CO1**
(a)	Given vectors $\vec{A} = 2\hat{a}_x + 2\hat{a}_y - \hat{a}_z$, $\vec{B} = \hat{a}_x - 3\hat{a}_y + 4\hat{a}_z$ and $\vec{C} = \hat{a}_x - \hat{a}_y + \hat{a}_z$ Determine (i) $\lvert \vec{A} + \vec{B} + \vec{C} \rvert$ (ii) $\vec{A} - \vec{B} + \vec{C}$ (iii) dot product of $\vec{B}$ and $\vec{C}$ (iv) $\vec{A} \times \vec{B} \bullet \vec{C}$	
(b)	Discuss different types of coordinate systems with suitable diagrams.	
(c)	State and explain: (i) Divergence theorem (ii) Stoke's theorem. Give the relevance of these theorems to electromagnetic problems	
Q2	(20 marks)	**CO1, CO2**
(a)	Explain Coulomb's law of electrostatics. Using this law, determine the electric field intensity due to the uniformly charged length of linear charge density ρ_L.	
(b)	Given the potential $V = \frac{10}{r^2} \sin\theta \cos\phi \, \hat{a}_r$ (i) Find the electric flux density $\vec{D}$ at $(2, \pi/2, 0)$. (ii) Calculate the work done in moving a $10\mu C$ charge from point $A(1, 30^0, 120^0)$ to $B(4, 90^0, 60^0)$	
(c)	Explain Biot-Savart's law. Determine the magnetic field due to a straight current carrying conductor of finite length with Biot-Savart's law.	
Q3	(20 marks)	**CO3, CO5**
(a)	What is Magnetic flux density? Discuss the different properties of magnetic flux lines.	
(b)	Explain Ampere's circuit law and subsequently find out Maxwell's third equation in both differential and integral form.	
(c)	What is magnetic boundary condition? Explain boundary conditions in two different magnetic media in detail.	
Q4	(20 marks)	**CO3, CO4**
(a)	What do you mean by an electromagnetic wave? Discuss the wavelength and time period of a wave moving in the positive x-axis.	
(b)	What is a waveguide? Discuss different types of waveguides with proper diagrams.	

(c)	Write the difference between waveguide and the transmission line. Explain TEM, TE, TM mode in rectangular waveguide.	
Q5	(20 marks)	**CO5, CO6**
(a)	What is the difference between a waveguide and a transmission line? Discuss different types of transmission line with suitable diagram.	
(b)	A distortionles line operating at 120 MHz has $= 20\,\Omega/m$, $L = 0.3\mu H/m$ and $C = 63\,pF/m$. Determine propagation constant(γ), phase velocity(V) and characteristics impedance (Z_0).	
(c)	What is a lossless transmission line? Determine the following parameters of a transmission line. (i) input impedance (ii) reflection coefficient (iii) VSWR	